

08 Shapley Additive Explanations

EP

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SHapley Additive exPlanations (SHAP)

SHAP values were used to interpret two classification models predicting species richness (“high” or “low”): a random forest, using `kernelshap`, and an XGBoost model using TreeSHAP. Both models were configured to explain predicted “high” richness. Because the XGBoost’s categorical predictors (ROFSW and ROFRS) were one-hot encoded during preprocessing, these were aggregated back into their original variables. This was to ensure fair, feature-for-feature comparison between models.

SHAP values for the RF are expressed on the probability scale, whereas XGBoost SHAP values are on the model’s log-odds scale. Their values, therefore, shouldn’t be directly compared but comparisons between relative feature importance and direction of effect can be made.

```
library(sf)
library(kernelshap)
library(shapviz)
library(dplyr)
library(tidymodels)
library(stringr)
library(patchwork)
library(targets)
library(ranger)

richness_fit_class <- readRDS("Data/richness_fit_class.rds")
richness_fit_xgb_ND <- readRDS("Data/richness_fit_xgb_ND.rds")
richness_recipe_xgb_ND <- readRDS("Data/richness_recipe_xgb_ND.rds")

predictors <- c("dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenpace")

grid_model_all <- st_read("Data/grid_model.gpkg")

## Reading layer 'grid_cells' from data source
## 'C:\Users\ellsp\OneDrive\Documents\1 Master's Project\RMDs\Data\grid_model.gpkg'
## using driver 'GPKG'
## Simple feature collection with 489 features and 21 fields
## Geometry type: POLYGON
## Dimension: XY
## Bounding box: xmin: 289277.6 ymin: 86966.9 xmax: 461661.2 ymax: 573772.4
## Projected CRS: OSGB36 / British National Grid
```

```
df <- grid_model_all %>%
  sf::st_drop_geometry()
```

Random Forest

```
# Extract model & set predictors
rf_model <- richness_fit_class %>%
  extract_fit_engine()
predictors <- df %>%
  dplyr::select(all_of(predictors))

# Set probability of "high"
predictor_function <- function(object, newdata) {
  predict(object, data = newdata)$predictions[, "high"]}

# Calc SHAP values
set.seed(399)

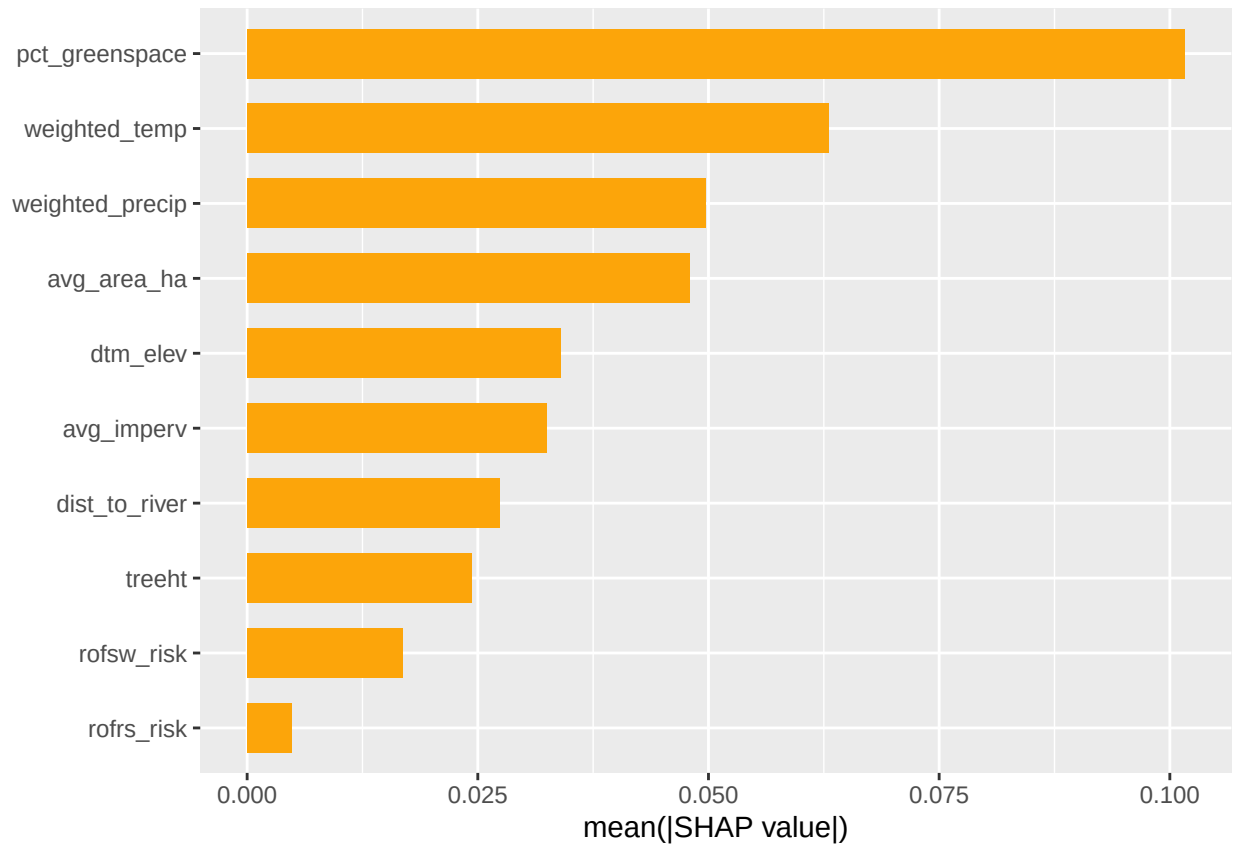
rf_shap <- kernelshap(
  object = rf_model,
  X = predictors,
  pred_fun = predictor_function)
```

```
## |
```

```
# Plot
sv_rf <- shapviz(rf_shap)

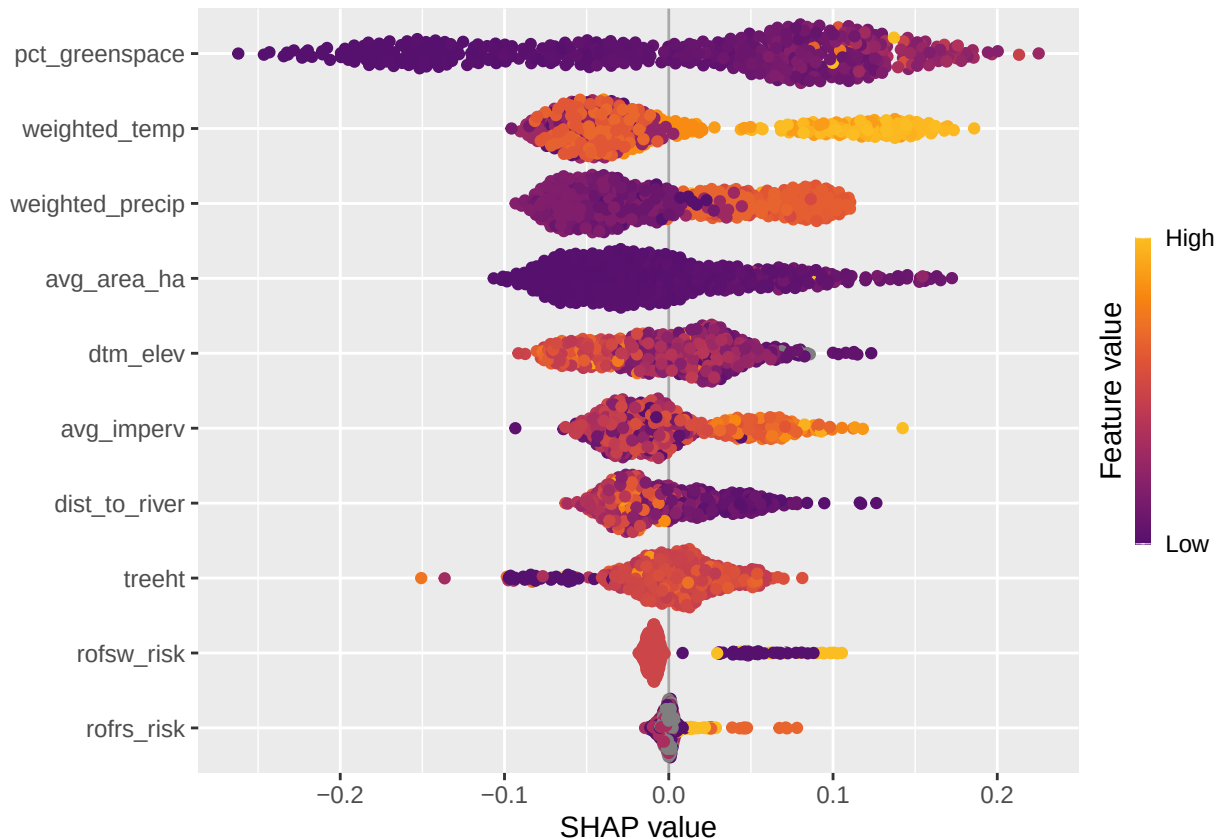
p1 <- sv_importance(sv_rf)
p2 <- sv_importance(sv_rf, kind = "beeswarm")

rfcombined <- p1 + p2
#ggsave("rf_shap.png", rfcombined, width = 12, height = 5, dpi = 300)
p1
```



This bar chart displays the SHAP values for the RF model, where the higher the SHAP value the greater the influence that predictor has on pushing the model towards predicting “high” richness. **pct_greenpace** is by far the most important predictor in this model, followed by **weighted_temp** and **weighted_precip**. Neither **ROFSW_risk** or **ROFRS_risk** are highly important in this model, but do provide some benefit.

p2



This beeswarm plot not only shows the importance of each predictor but also the direction of its effect and the associated feature value. For `pct_greenspace`, low values spread evenly across positive and negative SHAP values, while high values cluster on the positive side. This indicates high greenspace strongly increases the model prediction to high richness. `weighted_temp` showed a similar pattern, with high temperatures clustering positively and low-to-medium temperatures associated with low richness. The same relationship is seen for precipitation. Although smaller in overall impact, RoFSW showed a clear split, with low-risk clustering negatively and high-risk positively.

XGBoost

Because of the dummy step in the XGBoost model, a bake step is introduced to ensure the data is processed in the same way the model was. However, `kernelshap` is not needed here since the XGBoost package has TreeSHAP built in, which `shapviz` can use.

```
# Prepare predictor matrix
predictors <- c("dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenspace")

X_baked <- bake(richness_recipe_xgb_ND, new_data = df) %>%
  dplyr::select(-any_of("richness_class"))

# Set up SHAP
xgb_model <- richness_fit_xgb_ND %>%
  extract_fit_engine()
```

```

# Calc TreeSHAP values
sv_xgb <- shapviz(
  object = xgb_model,
  X_pred = as.matrix(X_baked),
  X = X_baked)

sv_xgb$S <- -sv_xgb$S
sv_xgb$baseline <- -sv_xgb$baseline

```

Rejoin ROFSW and ROFRS

Reform the original ROFSW and ROFRS categories for a beeswarm plot comparable with that of the RF.

```

# Map dummy cols to the correct category
shap_cols <- colnames(sv_xgb$S)
parent_vars <- c("rofrs_risk", "rofsw_risk")
col_map <- tibble(col = shap_cols) %>%
  mutate(parent = case_when(
    str_starts(col, "rofrs_risk") ~ "rofrs_risk",
    str_starts(col, "rofsw_risk") ~ "rofsw_risk",
    TRUE ~ col ))

# Sum SHAP values across dummy cols for each observation & aggregate
S <- as.data.frame(sv_xgb$S)

S_agg <- sapply(unique(col_map$parent), function(p) {
  cols <- col_map$col[col_map$parent == p]
  if (length(cols) == 1) {
    S[[cols]]} else {
    rowSums(S[cols])}})

S_agg <- as.data.frame(S_agg)
colnames(S_agg) <- unique(col_map$parent)

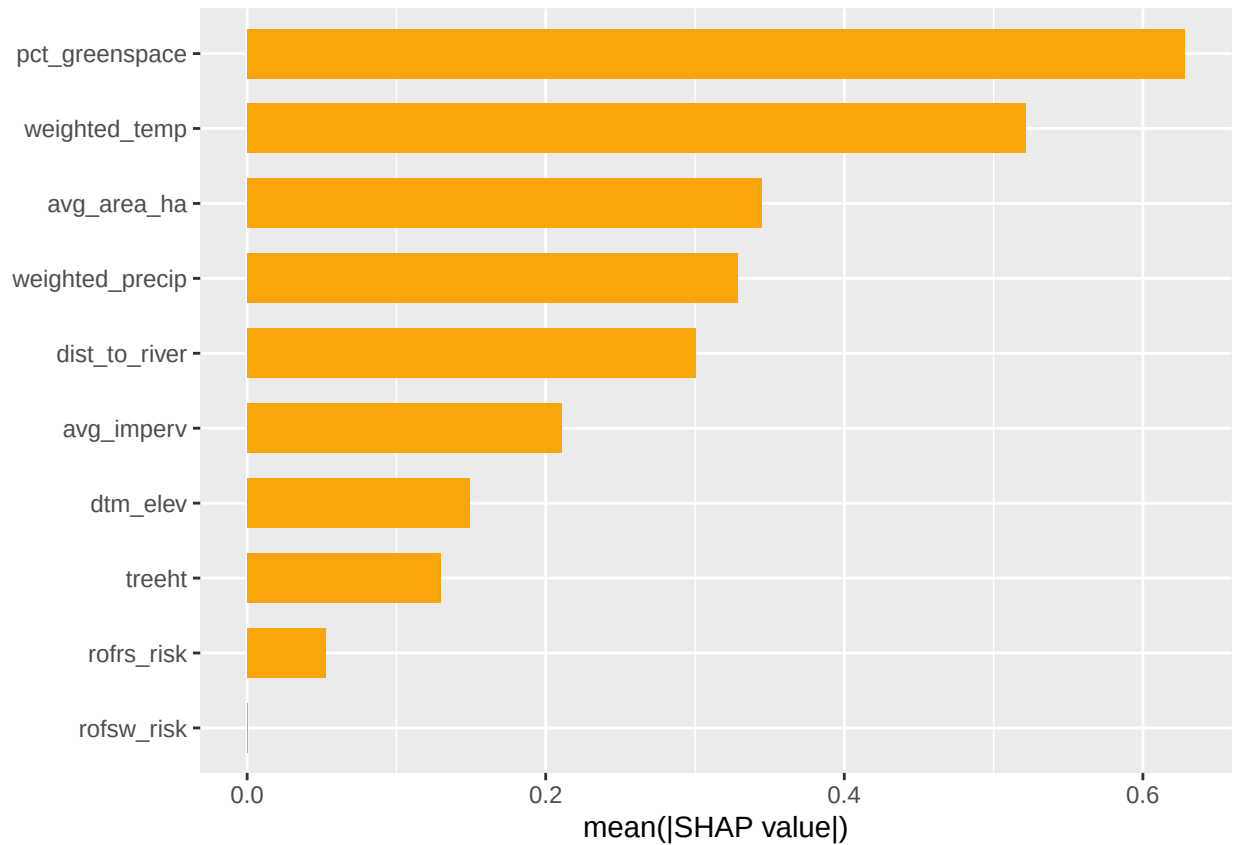
# Rebuild
X_agg <- df %>%
  dplyr::select(all_of(names(S_agg)[names(S_agg) %in% c("rofrs_risk", "rofsw_risk")]))
other_cols <- setdiff(colnames(S_agg), c("rofrs_risk", "rofsw_risk"))
X_agg <- cbind(X_baked[other_cols], X_agg)

# Build new shapviz object
sv_xgb_agg <- shapviz(
  object = as.matrix(S_agg),
  X = X_agg)

# Plot
p3 <- sv_importance(sv_xgb_agg)
p4 <- sv_importance(sv_xgb_agg, kind = "beeswarm")

xgbcombined <- p3 + p4
#ggsave("xgboost_shap.png", xgbcombined, width = 12, height = 5, dpi = 300)
p3

```



As seen in the RF model, the top feature is `pct_greenspace`, followed by `weighted_temp` and `weighted_precip`. As before, neither `ROFSW_risk` or `ROFRS_risk` are highly important in this model, but do provide some benefit.

p4

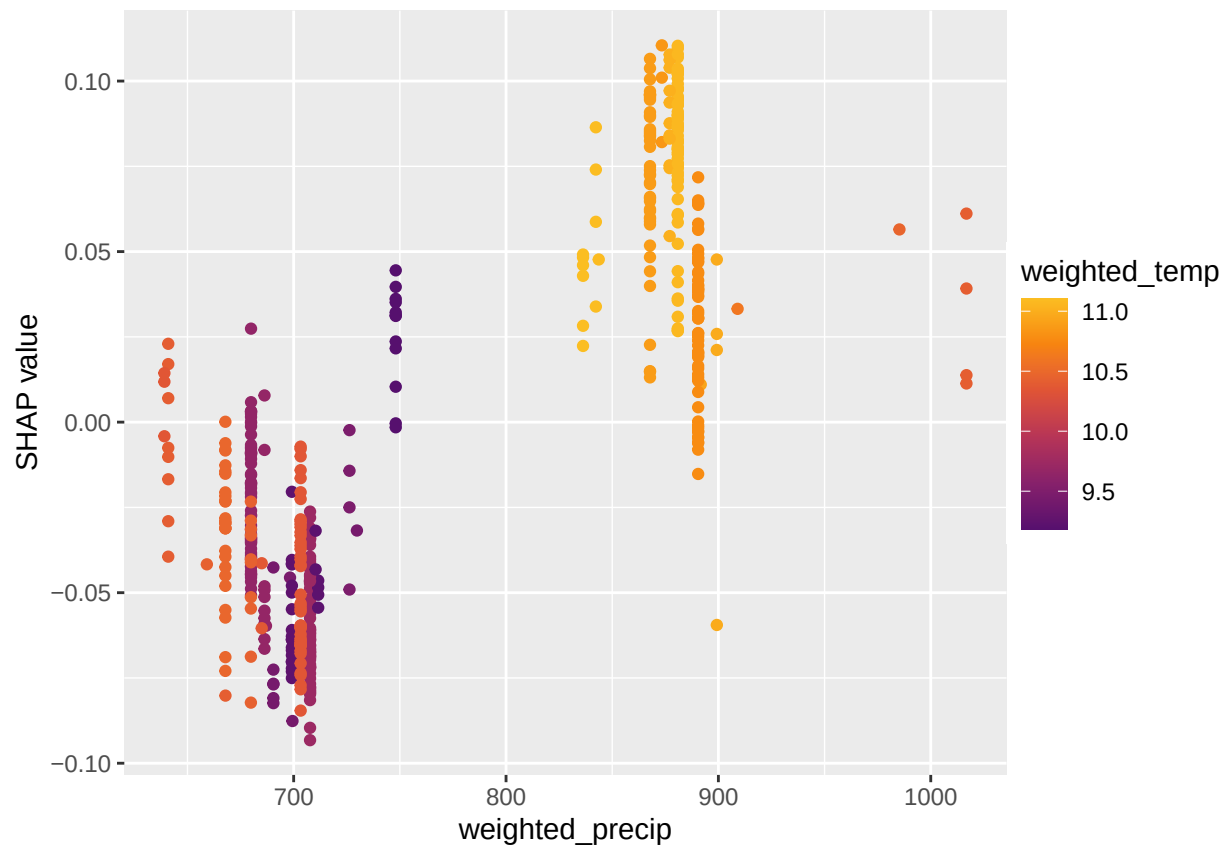


This beeswarm plot shows that the two models largely agree on both the important predictors and the direction of their effects. As in the RF model, `pct_greenspace` shows an even spread of low values but a clear cluster of high values with positive SHAP values. Both `weighted_temp` and `weighted_precip` again show higher values are associated with a higher likelihood of high richness being predicted. ROFSW risk shows the same pattern seen in the RF model.

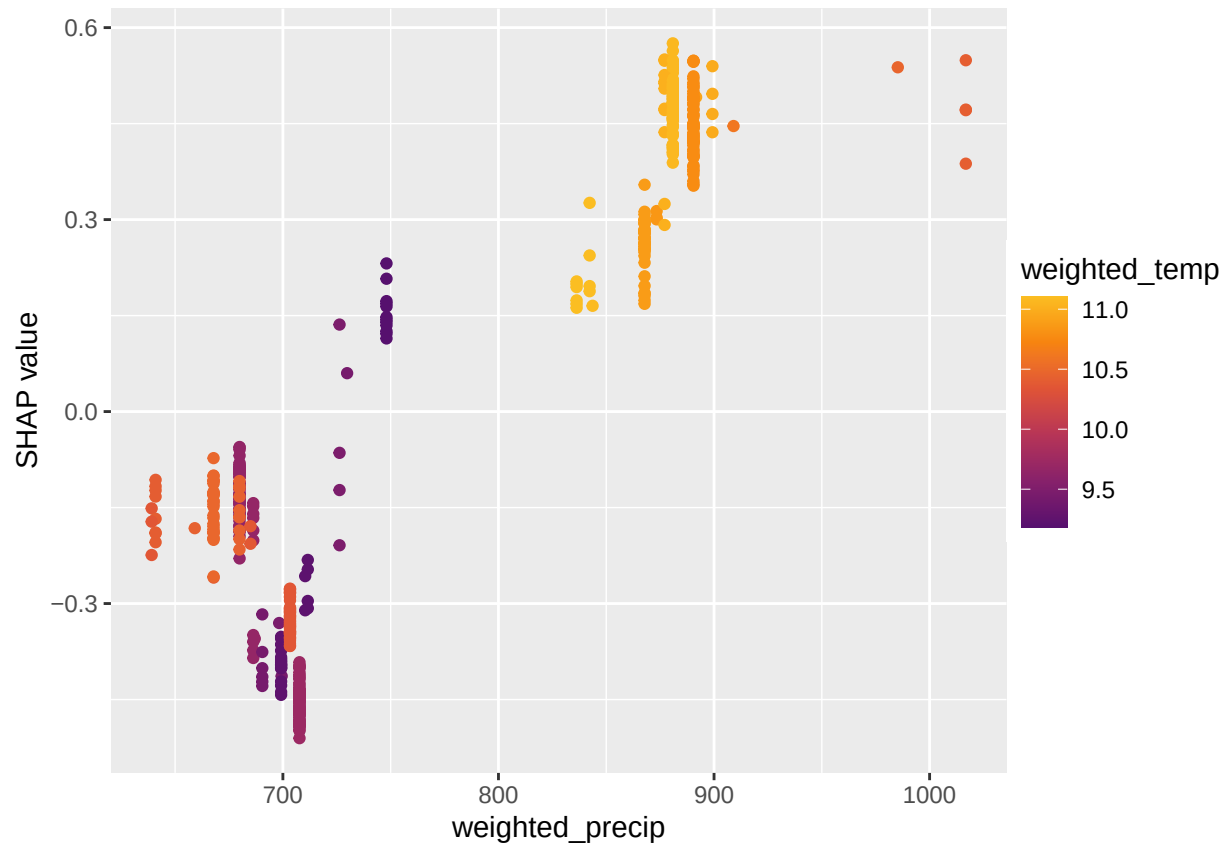
Precipitation and Temperature

```
p5 <- sv_dependence(sv_rf, "weighted_precip", color_var = "weighted_temp")
p6 <- sv_dependence(sv_xgb_agg, "weighted_precip", color_var = "weighted_temp")

precip_temp <- p5 + p6
p5
```



p6



```
#ggsave("precip_temp_shap.png", precip_temp, width = 12, height = 5, dpi = 300)
```

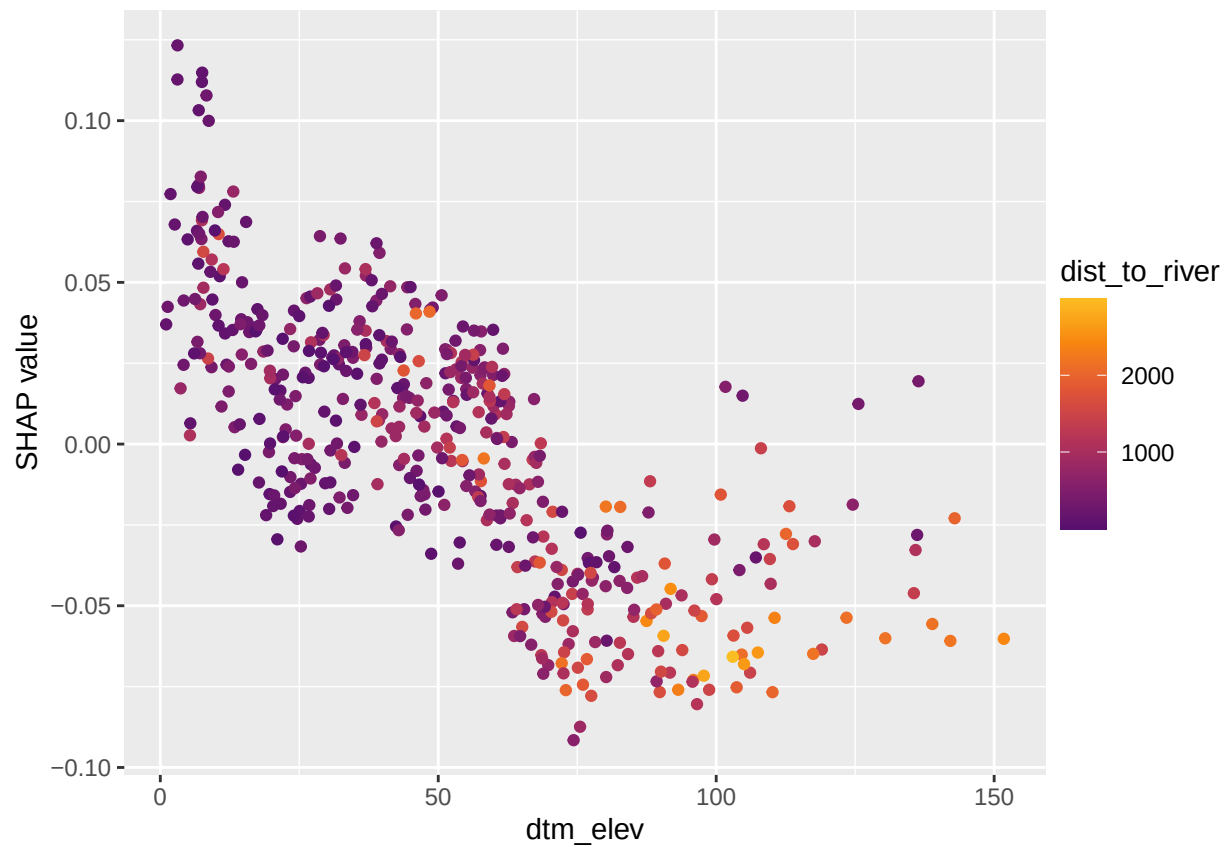
Since these two variables are correlated, a dependence plot was generated to examine their combined effect. Low temperature combined with low precipitation is associated with the model predicting low richness, whilst high values are associated with a predicted high richness. This relationship isn't linear and there appears to be a threshold that must be crossed before the prediction shifts from low to high richness.

Elevation vs Distance to River

```
p7 <- sv_dependence(sv_rf, "dtm_elev", color_var = "dist_to_river")
p8 <- sv_dependence(sv_xgb_agg, "dtm_elev", color_var = "dist_to_river")

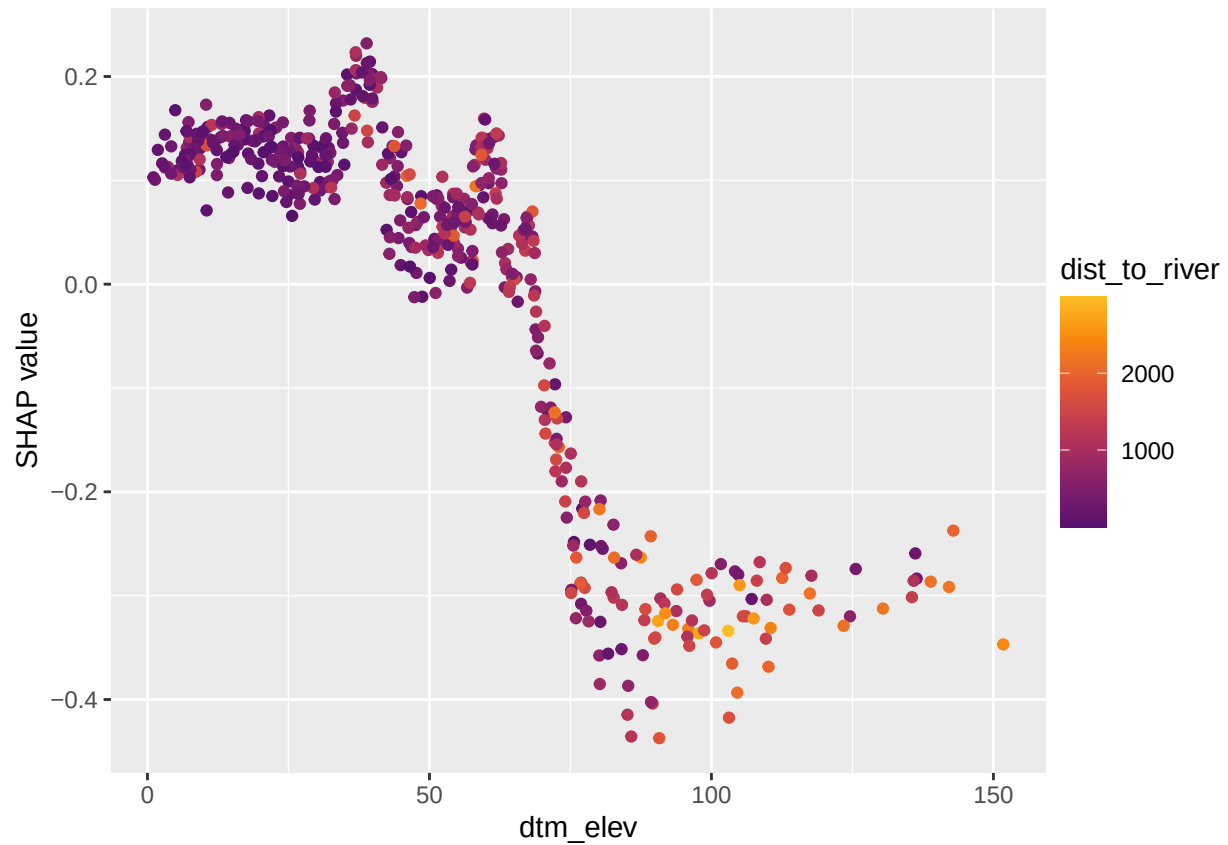
elevation_river <- p7 + p8
p7
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```



p8

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
#ggsave("elevation_v_river_shap.png", elevation_river, width = 12, height = 5, dpi = 300)
```

These dependence plots shows a more linear relationship between elevation and distance to the nearest river. Proximity to a river is associated with lower elevation and higher SHAP values, indicating a higher likelihood that high richness is predicted at low elevations, near rivers.